SUPPLEMENTARY GLENWOOD DECISION STUDY

Test the monitoring decision directly

Reject classifier policy promotion.

80.0%

classifier precision / known outcomes

8

known high windows captured

10

all proposed slots, including unknown

Direct high-window detection under a fixed daily capacity. This is a new hypothesis; prior North and East results remain unchanged.

`Baseline: 75.0% precision · 6 true alerts · 8 total slots`

`High outcomes: OOF 196/1923 (10.2%) → final 20/1510 (1.3%).`

Descriptive distribution shift; its cause is not established.

SOURCE · frozen Glenwood2024/2025 · METHOD · one declared decision-policy test
CONFIDENCE · retrospective, capacity-limited · USE · prospective validation; no realized benefit

ALLOCATION AND PAIRED UNCERTAINTY

Count every proposed slot

Observed replay counts	Baseline	Classifier
All eligible forecasts	1530	1530
Known future outcomes	1510	1510
All accepted slots	8	10
Accepted / unknown	0	0
True / false alerts	6 / 2	8 / 2
Missed high windows	14	12
Precision / recall	75.0% / 30.0%	80.0% / 40.0%

PAIRED PRECISION DIFFERENCE · 95% INTERVAL

5.000 [0.000, 13.333] pp

HIGH TARGET

126 FNU

Pooled development 90th percentile. Fixed before final opening.

PROMOTION GATES

PASS · Development gate

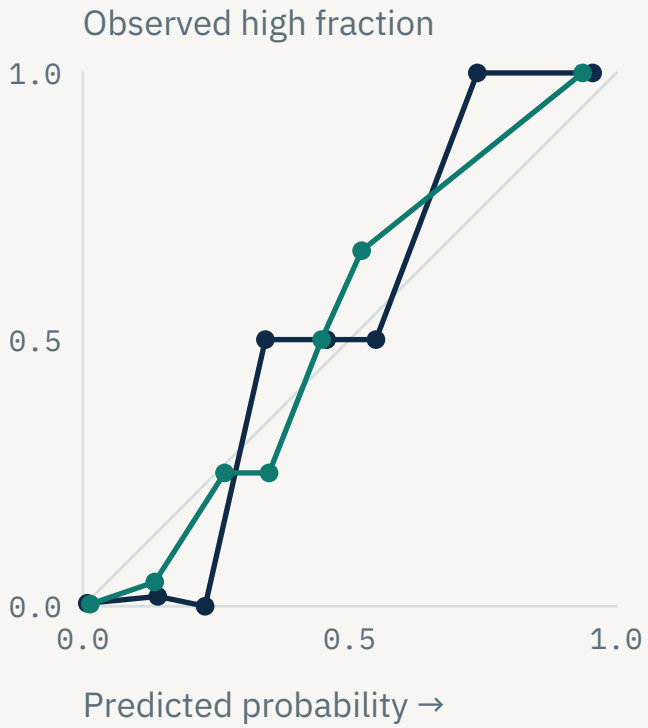
PASS · Capture ≥ baseline

FAIL · Slots ≤ baseline

FAIL · PPV interval above zero

SOURCE · all issued/known/unknown windows · METHOD · one slot/day before target QA
CONFIDENCE · shared evaluable calendar blocks · USE · compare capacity; costs remain hypothetical

Check the probabilities and both years



BRIER SCORE · LOWER IS BETTER

Classifier 0.007

Current logistic 0.009

Year	Known n	TP	Slots	Precision
2024	738	7	9	77.8%
2025	772	1	1	100.0%

Solid teal: selected · navy: current logistic

Fixed ten bins; empty bins omitted. Bin counts are in the evidence JSON. Raw-score baselines have no probabilities.

SOURCE · frozen probabilities / known outcomes · METHOD · fixed reliability bins
CONFIDENCE · sample support varies by bin/year · USE · probability audit, not post-holdout calibration

COMMERCIAL WORKFLOW AND ACCEPTANCE

Produce evidence for a human decision

Monitoring work-order draft

Watershed sponsor or water-operator monitoring lead

ACCEPTANCE BEFORE USE

Operator confirms access, feasible sampling schedule, laboratory quote and chain of custody before ordering.

Investment evidence register

Project sponsor and independent diligence reviewer

ACCEPTANCE BEFORE USE

Reviewer accepts each missing evidence item separately; forecast scores cannot approve financing.

Historical policy performance audit

Monitoring programme manager

ACCEPTANCE BEFORE USE

Choose a prospective pilot policy only after documented utility, field feasibility and live-data checks.

Historical replay ≠ live order · access, laboratory quote and chain of custody remain unverified.
No actual samples, committed buyer, avoided loss, ROI, credit or payout is established.

SOURCE · finance evidence ledger + policy audit · METHOD · explicit acceptance responsibilities
CONFIDENCE · workflow is implemented; commercial uptake unvalidated · USE · draft/review only